

Problems marked with a (*) are a bit more complex.

5.1. Derivatives.

- (a) Consider $f(x, y) := xy^2e^{-x^2-y^2}$ for $(x, y) \in \mathbb{R}^2$. Find all the critical points of f , that is all the points where the gradient of f vanishes. (The point will be given if and only if all the numerical values you find are correct... so check your computations twice).
- (b) We want to find a function $g \in C^1(\mathbb{R}^2)$ with the following directional derivative:

$$\partial_v g(x, y) = 2 \cos(x^2 y) x v_2 + \cos(x^2 y) v_1^2 y \quad \text{for all } (x, y) \text{ and } (v_1, v_2) \in \mathbb{R}^2.$$

Give an explicit example of such a function g or prove that a function with these properties cannot exist.

5.2. Lagrange multipliers. Consider the ball $U := \{(x, y, z) : x^2 + y^2 + z^2 < 1\}$ and the function

$$f(x, y, z) := 1 - x^2 - y^2 + 4x.$$

1. Motivate rigorously why f attains its absolute maximum and minimum in $\overline{U}$.
2. Find all the critical points of f which lie in the interior of U . Say whether $\sup_U f$ (or $\inf_U f$) is attained at some point in U .
3. Say whether $\max_{\partial U} f$ and $\min_{\partial U} f$ exist.
4. With the method of Lagrange multipliers, find the possible critical points for the constrained problem

$$\max\{f(x, y, z) : (x, y, z) \in \partial U\},$$

and same for min.

5. Among all the points you have found clarify at which points the maximum/minimum of f is attained.

5.3. Multiple choice. Mark all and only the statements which are true

- (a) As $|x| \rightarrow 0$, if $f(x) = x_2 x_1^2 + O(x_1^2)$, then $f(x) = O(|x|^2)$.
- (b) As $x \rightarrow 0$, if $f(x) = x_2 x_1^2 + O(x_1^3)$, then $f(x) = O(x_1^3)$.
- (c) As $|x| \rightarrow 0$, if $f(x) = x_2 x_1 + O(|x|^2)$, then f is differentiable at $x = 0$.
- (d) As $x \rightarrow 0$, if $f(x) = x_2 x_1 + O(|x|^2)$, then f is twice differentiable around $x = 0$.
- (e) As $|x| \rightarrow 0$, if $f(x) = x_2 x_1 + O(|x|^3)$, then f is twice differentiable around $x = 0$ and $\partial_{11} f(0) = 0$.

5.4. Computation of derivatives. For each of the following functions compute the directional derivative in the general direction v at a general point x , that is $\partial_v f(x)$, where applicable.

1. $x_1/x_2, e^{-x_1/x_2}$ for $x \in \mathbb{R}^2 \setminus \{x_2 = 0\}$. (Also: Can they be extended continuously or differentiably across $\{x_2 = 0\}$?)
2. $e^{x_1 x_2} \sin(x_1 + x_2^2), \frac{x_1^2}{x_1^2 + x_2^2}, \frac{x_1^2 x_2}{x_1^2 + x_2^2}, \frac{x_1^2 x_2^2}{x_1^2 + x_2^2}$ for $x \in \mathbb{R}^2 \setminus \{0\}$. (Also: Can they be extended continuously or differentiably across the origin?)
3. $|x|, |x|^\alpha, x/|x|, g(|x|), x \cdot e, g(x \cdot e)$, for $\alpha \in \mathbb{R}, e \in \mathbb{R}^n$ and $g \in C^1(\mathbb{R})$.
4. $ax, x^T, \text{Tr}(x), x^2, x^3, \text{Tr}(x^2)$ for $x, a \in \mathbb{R}^{n \times n}$,
5. (*) x^{-1}, x^{-2} for $x \in \mathbb{R}^{n \times n}$ with $\det x \neq 0$.

5.5. Multiple choice. Assume $g \in C^\infty(\mathbb{R})$ and $f \in C^\infty(\mathbb{R}^n)$ is such that

$$f(x) = x_1 + x_2 x_1 + O(|x|^4) \text{ as } x \rightarrow 0.$$

We want to compute the Taylor expansion of $(g \circ f)$ at $x = 0$ of the highest possible order with the information we have.

We can ask for the value $g^{(k)}(t)$ for arbitrary $k \in \mathbb{N}, t \in \mathbb{R}$, but we have to pay 5 CHF each time.

For a general g , which is the degree of the best (i.e., highest-order) Taylor polynomial we can compute? How expensive will it be to compute it?

- (a) degree 3 and 20 CHF
- (b) degree 2 and 15 CHF
- (c) degree 2 and 10 CHF
- (d) degree 3 and 15 CHF

5.6. Polynomials. Let $P, Q \in \mathbb{R}[x_1, \dots, x_n]$ be polynomials of n variables, assume that there are positive numbers M, σ such that

$$|P(x) - Q(x)| \leq M|x|^\sigma \text{ for all } |x| \leq 1.$$

Show that P and Q have the same coefficients of order smaller than σ . That is, if we write

$$P(X) = \sum_{\alpha \in \mathbb{N}^n} a_\alpha X^\alpha, \quad Q(X) = \sum_{\alpha \in \mathbb{N}^n} b_\alpha X^\alpha,$$

then $a_\alpha = b_\alpha$ for all $|\alpha| < \sigma$.

5.7. Chain rule for a curve, or: a car on Earth. Let $R > 0$ and $\omega \in \mathbb{R}$ be constants. Let $\sigma : I \subset \mathbb{R} \rightarrow \mathbb{R}^2, \sigma(t) = (a(t), b(t))$, which we interpret as a curve in a “map” representing the longitude and latitude of a car at time t .

Define the “rotating” spherical coordinates $F : \mathbb{R}^2 \times \mathbb{R} \rightarrow \mathbb{R}^3$,

$$F((x, y), t) = (R \cos y \cos(\omega t + x), R \cos y \sin(\omega t + x), R \sin y),$$

representing what a point on the map looks like on Earth from a satellite (thus we consider the Earth as a rotating sphere with speed given by ω , and we forget about translation around the Sun) at time t .

Consider $r : \mathbb{R} \rightarrow \mathbb{R}^3$, $r(t) = F(\sigma(t), t)$, which is the position of the car at time t .

1. Compute the acceleration $r''(t)$ using the chain rule.
2. Assume that the acceleration is always proportional to the radial vector $r(t)$, i.e.

$$r''(t) = \lambda(t) r(t)$$

for some scalar function $\lambda(t)$. Show that a and b satisfy

$$b'' + \sin b \cos b (\omega + a')^2 = 0,$$

and

$$\cos b a'' - 2 \sin b (\omega + a') b' = 0.$$

Equivalently, away from the poles ($\cos b \neq 0$),

$$a'' - 2 \tan b (\omega + a') b' = 0.$$

Hints:

5.3 Revise Corollary 10.45.

5.4 In order to prove that a function **is** differentiable at 0 you have two tools

- the Definition of differentiability;
- the sufficient condition of Theorem 10.16.

In order to prove that a function **is not** differentiable at 0 you have two tools

- Show that the necessary condition of Proposition 10.14 does not hold;
- Show that for some C^1 curve $\gamma: (-\delta, \delta) \rightarrow \mathbb{R}^n$ the function $(f \circ \gamma)(t)$ is **not** differentiable at $t = 0$. This could be simpler as you have to study a function of only one variable.

5.4.3 These functions are C^∞ so try to compute them on $x + tv$ and find an expansion in powers of t ... If you find something it must be the Taylor polynomial by Corollary 10.45. From the Taylor polynomial it is immediate to read off the derivatives. For example

$$(x + tv)^2 = (x + tv)(x + tv) = x^2 + txv + tvx + t^2 = x^2 + (xv + vx)t + O(t^2)...$$